

LAB EXPERIMENT-1

- AIM:** i) To design and implement an 8×1 Multiplexer (MUX) using Logisim, and verify its operation using different input combinations.
ii) To design and implement an 8×1 MUX using 2x1 MUX
iii) To design and implement 8x3 encoder

THEORY:-

A Multiplexer (MUX) is a combinational circuit that selects one input from multiple inputs and forwards it to a single output line.

8×1 Multiplexer

- * Number of inputs = 8 (A₀, A₁, A₂, ..., A₇)
- * Number of select lines = 3 (S₂, S₁, S₀)
- * Output = Y

INPUTS			Output
S ₂	S ₁	S ₀	Y
0	0	0	A ₀
0	0	1	A ₁
0	1	0	A ₂
0	1	1	A ₃
1	0	0	A ₄
1	0	1	A ₅
1	1	0	A ₆
1	1	1	A ₇

$$Y = A_0 S_2' S_1' S_0' + A_1 S_2' S_1' S_0 + A_2 S_2' S_1 S_0' + A_3 S_2' S_1 S_0 \\ + A_4 S_2 S_1' S_0' + A_5 S_2 S_1' S_0 + A_6 S_2 S_1 S_0' + A_7 S_2 S_1 S_0$$

An encoder is a combinational circuit that converts 2^n input lines into n output lines, producing a binary representation of the active input.

In an 8×3 encoder:

- There are 8 inputs (D0-D7)
- There are 3 outputs (X, Y, Z)

Only one input is assumed HIGH at a time.

Input								Output		
D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇	X	Y	Z
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

$$X = D_1 + D_3 + D_5 + D_7$$

$$Y = D_2 + D_3 + D_6 + D_7$$

$$Z = D_4 + D_5 + D_6 + D_7$$

PROCEDURE FOR MUX:-

Step 1: Open Logisim

- Launch Logisim
- Create a new circuit

Step 2: Place Components

- From “Plexers” select Multiplexer
- Set:
 - Data Bits = 1
 - Select Bits = 3

Step 3: Add Inputs and Output

Add:

- 8 input pins → A0 to A7
- 3 select pins → S2, S1, S0
- 1 output pin → Y

Step 4: Connect Circuit

- Connect all inputs to the multiplexer
- Connect select lines properly
- Connect output to output pin

Step 5: Simulation

- Toggle input values (0/1)
- Change select lines
- Observe output

PROCEDURE FOR ENCODER:-**Step 1: Open Logisim**

- Launch Logisim
- Create a new circuit

Step 2: Place Components

- From “Wiring” select Input Pin (place 8)
- From “Wiring” select Output Pin (place 3)
- From “Gates” select OR Gate (place 3)
- Set each OR gate to 4 inputs

Step 3: Add Inputs and Outputs

Add:

- 8 input pins → D0 to D7
- 3 output pins → X, Y, Z

Step 4: Connect Circuit

- Connect D1, D3, D5, D7 → OR gate → Y0
- Connect D2, D3, D6, D7 → OR gate → Y1
- Connect D4, D5, D6, D7 → OR gate → Y2

Step 5: Simulation

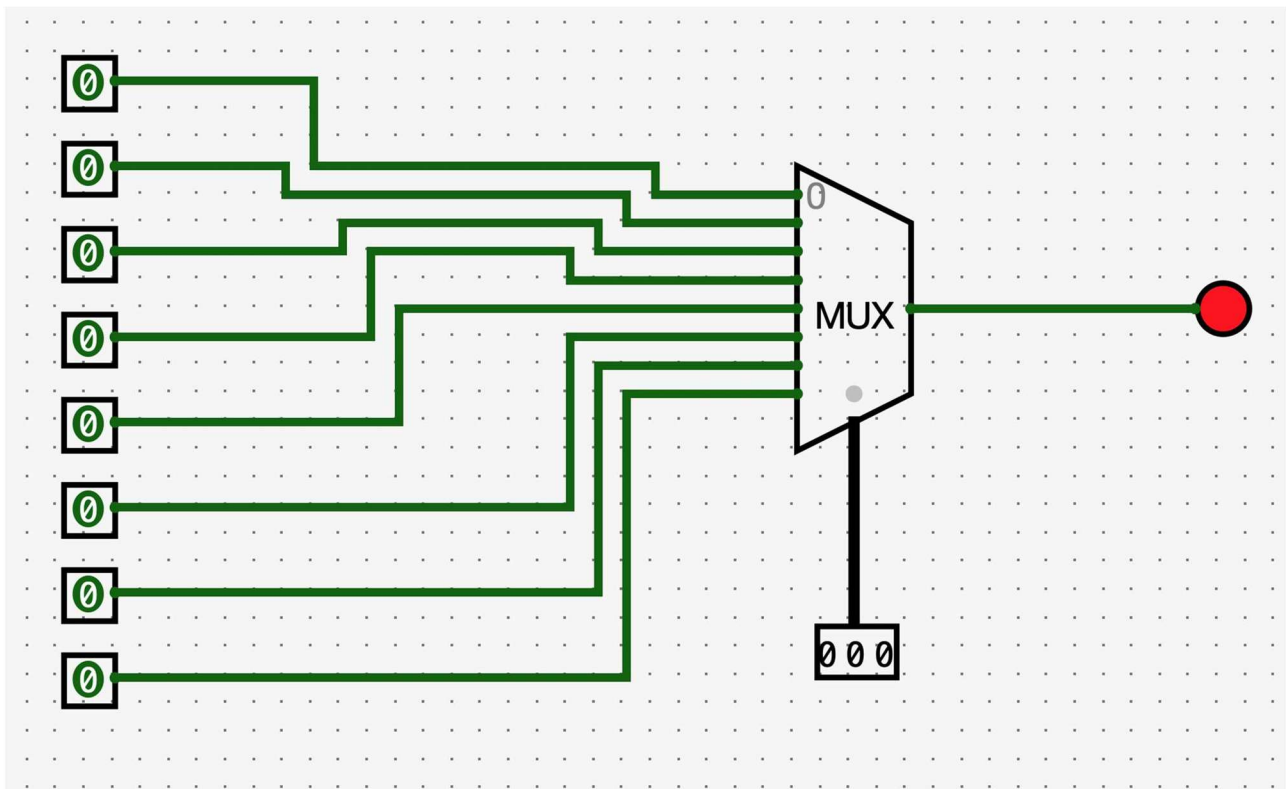
- Toggle one input at a time (0/1)
- Observe output changes

Step 6: Verification

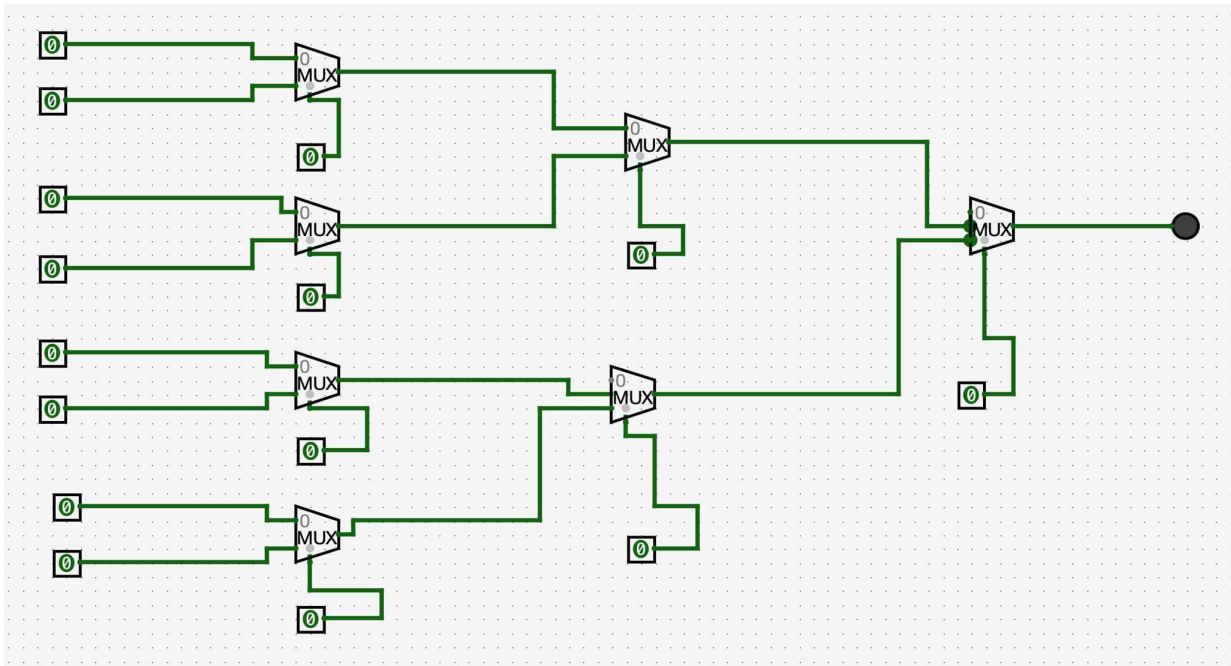
- Check outputs match encoder truth table
- Ensure correct binary output is obtained

OBSERVATION:-

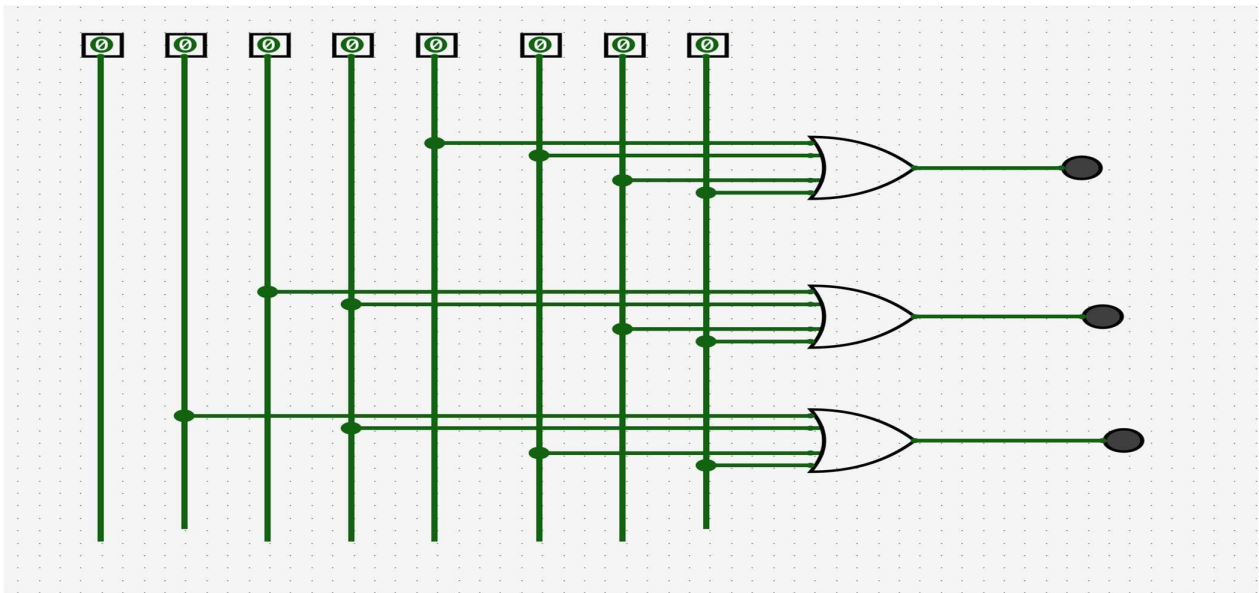
1)8x1 Multiplexer :



2) 8x2 Multiplexer :



3) Encoder :



On applying different input combinations and select lines, the 8×1 multiplexer was observed to pass only the selected input to the output while all other inputs were ignored. In the 8×2 multiplexer, it was observed that the circuit, built using smaller multiplexers, produced corresponding two-bit outputs based on the select lines. For the encoder circuit, it was observed that when a single input line was active, the output generated the equivalent binary

code. The outputs changed accordingly with changes in inputs and select lines, confirming the functional behavior of each circuit during testing.

Conclusion :

The implementation of the 8×1 multiplexer, 8×2 multiplexer, and encoder circuits was successfully completed and tested. Each circuit produced the expected output for different input combinations, confirming correct working and proper design. The experiment helped in understanding how multiplexers can efficiently select data using control signals and how encoders convert multiple inputs into binary form. Overall, the objectives of the experiment were achieved, and the practical knowledge of digital logic design was strengthened